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PUBLICATION LIST

PATENTS

- [P1] Reznik, L., & **Chuprov, S.** “Federated Learning with A Compromised Unit Exclusion from Receiving Global Model Updates”.
Provisional application filed on January 13, 2023. Converted to non-provisional
- [P2] Reznik, L., & **Chuprov, S.** “Network Adjustment based on Machine Learning End System Performance Monitoring Feedback”.
Provisional application filed on September 14, 2022. Converted to non-provisional

BOOKS & BOOK CHAPTERS

- [B1] **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2025) “mLINK: Machine Learning Integration with Network and Knowledge” in Metacognitive Artificial Intelligence, eds. Shakarian, P. & Wei, H. Cambridge University Press, September 2025, Chapter 16, pp. 264-275. doi: 10.1017/9781009522472. [URL](#)

REPRINTS

- [R1] **Chuprov, S.**, Belyaev, P., Gataullin, R., Reznik, L., Neverov, E., & Viksnin, I. (2024) “Robust Autonomous Vehicle Computer-Vision-Based Localization in Challenging Environmental Conditions” in Applied Sciences. 2024, pp. 34-48., doi: doi.org/10.3390/books978-3-7258-2184-6. *Reprint of the Special Issue “Challenges in the Guidance, Navigation and Control of Autonomous and Transport Vehicles” that was published in Applied Sciences in 2023.* [URL](#)

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- [J1] Korobeinikov, D., Zatsarenko, R., **Chuprov, S.**, Barea, A., & Reznik, L. (2026). “InteFL Framework: Optimizing Federated Learning with Metacognition for Application Design and Deployment” in IEEE Intelligent Systems. 2026. doi: 10.1109/MIS.2026.3658072. [URL](#). **Q1**. *Published in Special Issue*
- [J2] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2025). “KISS: Knowledge Integration System Service for ML End Attacks Detection and Classification” in IEEE Transactions on Dependable and Secure Computing. 2025, pp. 1-12. doi: 10.1109/TDSC.2025.3553807. [URL](#). **Q1**.
- [J3] **Chuprov, S.**, Zatsarenko, R., Reznik, L., & Khokhlov, I. (2024). “Data Quality Based Intelligent Instrument Selection with Security Integration” in ACM Journal of Data and Information Quality. 2024, vol. 16, no. 3, pp. 1-24. doi: 10.1145/3695770. [URL](#). **Q2**.
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- [J5] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev, E. (2023). “Review and Comparison of Lightweight Modifications of the AES Cipher for a Network of Low-Power Devices” in Automatic Control and Computer Sciences. 2022, no. 56, pp. 994-1006. doi: 10.3103/S0146411622080028. [URL](#)
- [J6] Melnikov, T., **Chuprov, S.**, Lazarev, E., Gataullin, R., & Viksnin, I. (2022). “Improving Reputation and Trust-Based Approach with Reliability Indicators for Autonomous Vehicles Intergroup Communication” in Tomsk State University Journal of Control and Computer Science. 2022, no. 61, pp. 108-117. [URL](#)
- [J7] Marinenkov, E., **Chuprov, S.**, Tursukov, N., Kim, I., & Viksnin, I. (2022). “Study on Destructive Informational Impact in Unmanned Aerial Vehicles Intergroup Communication” in Symmetry. 2022. Vol. 14, no. 8, pp. 1-18. [URL](#). **Q2**. *Published in Special Issue*
- [J8] Viksnin, I., Marinenkov, E., & **Chuprov, S.** (2022). “A Game Theory approach for communication security and safety assurance in cyber-physical systems with Reputation and Trust-based mechanisms” in Scientific and Technical Journal of Information Technologies, Mechanics and Optics, vol. 22, no. 1, pp. 47-59. doi: 10.17586/2226-1494-2022-22-1-47-59. [URL](#)

- [J9] Khokhlov, I., Reznik, L., & **Chuprov, S.** (2020). “Framework for Integral Data Quality and Security Evaluation in Smartphones” in IEEE Systems Journal, vol. 15, no. 2, pp. 2058-2065, doi: 10.1109/JSYST.2020.2985343. [URL](#). **Q1**
- [J10] **Chuprov, S.**, Viksnin, I., Kim, I., Tursukov, N., & Nedosekin, G. (2020). Empirical Study on Discrete Modeling of Urban Intersection Management System. International Journal of Embedded and Real-Time Communication Systems (IJERTCS), vol. 11(2), pp. 16-38, doi: 10.4018/IJERTCS.2020040102. [URL](#)
- [J11] **Chuprov, S.**, Viksnin, I., Kim, I., Marinenkov, E., Usova, M., Lazarev, E., Melnikov, T., & Zakoldaev, D. (2019). Reputation and Trust Approach for Security and Safety Assurance in Intersection Management System. Energies, 12(23), 4527, doi: 10.3390/en12234527. [URL](#). **Q1**

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- [C1] **Chuprov, S.**, Lange, R. D., Reznik, L., Shakarian, P., Zatsarenko, R., & Korobeinikov, D. (2026). “Position: Artificial Intelligence Needs Meta Intelligence — the Case for Metacognitive AI” in Proceedings of the 43rd International Conference on Machine Learning (ICML 2026). *Accepted*. [URL](#)
- [C2] Zatsarenko, R., Korobeinikov, D., **Chuprov, S.**, & Reznik, L. (2026). “Computationally Efficient Anomaly Detection and Exclusion for Practical and Robust Federated Learning” in IEEE/IFIP International Conference on Dependable Systems and Networks (DSN 2026). *To appear*.
- [C3] Peña, A. & **Chuprov, S.** (2026). “Improving PID-Based Aggregation in Federated Learning for Consumer Applications” in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366291. [URL](#)
- [C4] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2026). “How to Improve Federated Learning in Consumer Applications? Detect and Exclude Bad Clients” in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366313. [URL](#)
- [C5] Peredo, A. & **Chuprov, S.** (2026). “Improving Generalization with Harmonic Aggregation in Personalized Federated Learning” in 2026 IEEE 23rd Consumer Communications & Networking Conference (CCNC). 2026. doi: 10.1109/CCNC65079.2026.11366390. [URL](#)
- [C6] Zatsarenko, R., **Chuprov, S.**, Korobeinikov, D., Reznik, L., & Peña, A. (2025). “MADE-PI: Framework for Effective Anomaly Detection in Federated Learning Applications” in 2025 International Conference on Machine Learning and Applications (ICMLA). 2025, pp. 446-451, doi: 10.1109/ICMLA66185.2025.00067. [URL](#)
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- [C8] Chan, E. E. N. & **Chuprov, S.** (2025). “Bayesian Optimization of PID Coefficients for Robust Federated Learning” in TACCSTER 2025 Proceedings. 2025. doi: 10.26153/tsw/62178. [URL](#)
- [C9] Hong, L., **Chuprov, S.**, & Wan, Y. (2025) “Mitigating Health Misinformation Through Human-AI Collaboration” in AMCIS 2025 Proceedings. [URL](#)
- [C10] **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2025) “No trust, no learning? Improving Federated Learning Security and Robustness with Reputation and Trust” in Data Science Workshop “From Theory to Practice”. [URL](#)
- [C11] Korobeinikov, D., **Chuprov, S.**, & Reznik, L., (2024). “Towards More Robust Federated Learning with Medical Imaging Model Anomaly Detection” in 2024 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2024, pp. 1-4. [URL](#)
- [C12] Narayanan, A., **Chuprov, S.**, Reznik, L., Zatsarenko, R., & Korobeinikov, D. “Intelligent Soccer Event Detection and Highlights Generation with Broadcast Cues Integration” in 23rd International Conference on Machine Learning and Applications (ICMLA’24), Miami, FL, USA, 2024, pp. 554-559, doi: 10.1109/ICMLA61862.2024.00081. [URL](#)

- [C13] Patel, H., **Chuprov, S.**, Korobeinikov, D., Zatsarenko, R., & Reznik, L. (2024). "Improving Federated Learning Security with Trust Evaluation to Detect Adversarial Attacks" in 19th Annual Symposium on Information Assurance (ASIA' 24), 2024, pp. 37-43. [URL](#)
- [C14] Korobeinikov, D., **Chuprov, S.**, Zatsarenko, R., & Reznik, L., (2024). "Federated Learning Robustness on Real World Data in Intelligent Transportation Systems" in 19th Annual Symposium on Information Assurance (ASIA' 24), 2024, pp. 30-36. [URL](#)
- [C15] **Chuprov, S.**, Zatsarenko, R., Korobeinikov, D., & Reznik, L. (2024). "Robust Training on the Edge: Federated vs. Transfer Learning for Computer Vision in Intelligent Transportation Systems" in 2024 IEEE World AI IoT Congress (AIIoT), Seattle, WA, USA, 2024, pp. 172-178, doi: 10.1109/AIIoT61789.2024.10578970. [URL](#)
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- [C19] **Chuprov, S.**, Mahajan, S., Zatsarenko, R., Reznik, L. & Ruchkan, A. (2023). "Are Industrial ML Image Classifiers Robust to Withstand Adversarial Attacks on Videos?" in 2023 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2023, pp. 1-4, doi: 10.1109/WNYISPW60588.2023.10349595. [URL](#)
- [C20] Zatsarenko, R., **Chuprov, S.**, Marathe, C.A., Hyland, M., & Reznik, L. (2023). "Are Industrial ML Image Classifiers Robust to Data Affected by Network QoS Degradation?" in 2023 IEEE Western New York Image and Signal Processing Workshop (WNYISPW), 2023, pp. 1-4, doi: 10.1109/WNYISPW60588.2023.10349560. [URL](#)
- [C21] Neverov, E., Viksnin, I., & **Chuprov, S.** (2023) "The Research of AutoML Methods in the Task of Wave Data Classification" in 2023 XXVI International Conference on Soft Computing and Measurements (SCM), 2023, pp. 156-158, doi: 10.1109/SCM58628.2023.10159058. [URL](#)
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- [C23] **Chuprov, S.**, Bhatt, K.M., & Reznik, L. (2023). "Federated Learning for Robust Computer Vision in Intelligent Transportation Systems" in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 26-27, doi: 10.1109/CAI54212.2023.00019. [URL](#)
- [C24] **Chuprov, S.**, Memon, M., & Reznik, L. (2023). "Federated Learning with Trust Evaluation for Industrial Applications" in 2023 IEEE Conference on Artificial Intelligence (CAI), Santa Clara, CA, USA, 2023, pp. 347-348, doi: 10.1109/CAI54212.2023.00153. [URL](#)
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- [C27] **Chuprov, S.**, Khokhlov, I., Reznik, L., & Manghi, K. (2022). "Multi-Modal Sensor Selection with Genetic Algorithms" in 2022 IEEE Sensors, 2022, pp. 1-4, doi: 10.1109/SENSORS52175.2022.9967296. [URL](#)

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- [C29] **Chuprov, S.**, Khokhlov, I., Reznik, L., & Shetty, S. (2022). "Influence of Transfer Learning on Machine Learning Systems Robustness to Data Quality Degradation" in 2022 International Joint Conference on Neural Networks (IJCNN 2022), 2022, pp. 1-8, doi: 10.1109/IJCNN55064.2022.9892247. [URL](#)
- [C30] **Chuprov, S.**, Reznik, L., Obeid, A., & Shetty, S. (2022). "How Degrading Network Conditions Influence Machine Learning End Systems Performance?" in IEEE INFOCOM 2022 - IEEE Conference on Computer Communications Workshops (INFOCOM WKSHPS), 2022, pp. 1-6, doi: 10.1109/INFOCOMWKSHPS54753.2022.9798388. [URL](#)
- [C31] **Chuprov, S.**, Gataullin, R., Neverov, E., Belyaev, P., Kim, I., & Viksnin, I. (2022). "Police Office Model Performance and Security Evaluation in a Simulated Group of Mobile Robots" in The 5th International Conference on Future Networks & Distributed Systems (ICFNDS 2021). Association for Computing Machinery, New York, NY, USA, pp. 606–615, doi: 10.1145/3508072.3508195. [URL](#)
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- [C34] Lyakhovenko, Y., Viksnin, I., & **Chuprov, S.** (2021). "Integrating Smart Contracts into Smart Factory Elements' Informational Interaction Model" in CEUR Workshop Proceedings, Vol. 2893, pp. 1-6. [URL](#)
- [C35] Usova, M., Viksnin, I., & **Chuprov, S.** (2021). "Informational Messages and Space Models Application in Smart Factory Concept" in CEUR Workshop Proceedings, Vol. 2893, pp. 1-8. [URL](#)
- [C36] Khanh, T.D., Komarov, I., Don, L.D., Iureva, R., & **Chuprov, S.** (2020). "TRA: Effective Authentication Mechanism For Swarms Of Unmanned Aerial Vehicles" in 2020 IEEE Symposium Series on Computational Intelligence, pp. 1852-1858, doi: 10.1109/SSCI47803.2020.9308140. [URL](#)
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- [C38] Usova, M., **Chuprov, S.**, & Viksnin, I. (2020). "Informational Space and Messages Interaction Models for Smart Factory Concept" in 2020 IEEE International Workshop on Metrology for Industry 4.0 & IoT, pp. 617-621, doi: 10.1109/MetroInd4.0IoT48571.2020.9138292. [URL](#)
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- [C45] Marinenkov, E., **Chuprov, S.**, Viksnin, I., & Kim, I. (2019). “Empirical Study on Trust, Reputation, and Game Theory Approach to Secure Communication in a Group of Unmanned Vehicles” in CEUR Workshop Proceedings, vol. 2590, pp. 1-12. [URL](#)
- [C46] Usova, M., **Chuprov, S.**, Viksnin, I., & Baranova, O. (2019). “Model of Secure Informational Messages for Ensuring Informational Interaction in Smart Factory” in CEUR Workshop Proceedings, vol. 2590, pp. 1-8. [URL](#)
- [C47] Viksnin, I., Lyakhovenko, J., Tursukov, N., **Chuprov, S.**, & Sozinova, E. (2019). “Empirical Study on Modeling of People Behavior in Emergency” in CEUR Workshop Proceedings, vol. 2590, pp. 1-8. [URL](#)
- [C48] Kim, I., Matos-Carvalho, J. P., Viksnin, I., Campos, L. M., Fonseca, J. M., Mora, A., & **Chuprov, S.** (2019). “Use of Particle Swarm Optimization in Terrain Classification based on UAV Downwash” in 2019 IEEE Congress on Evolutionary Computation (CEC), pp. 604-610, doi: 10.1109/CEC.2019.8790031. [URL](#)
- [C49] Viksnin, I., **Chuprov, S.**, Usova, M., & Zakoldaev, D. (2019). “Police Office Model for Multi-agent Robotic Systems” in IOP Conference Series: Materials Science and Engineering, vol. 497, no. 1, p. 012036, doi: 10.1088/1757-899X/497/1/012036. [URL](#)

DISSERTATIONS

- [D1] **Chuprov, S.** (2024) “Robust Machine Learning Under Vulnerable Cyberinfrastructure and Varying Data Quality”, Ph.D. dissertation, B. Thomas Golisano College of Computing and Information Sciences, Rochester Institute of Technology, Rochester, NY, USA, 2024. [URL](#)
- [D2] **Chuprov, S.** (2019) “Assurance of Information in a Mobile Robotic System with Decentralized Control”, Master’s dissertation, Department of Secure Information Technologies, ITMO University, Saint Petersburg, Russia, 2019.

OTHER PUBLICATIONS AND PRESENTATIONS

- [O1] Fonseca, P., & **Chuprov, S.** (2025). “SPAM Email Detection App Using Machine Learning Techniques” in *UTRGV 2025 STEM Research Conference*. Presented May 2, 2025. — *Best Doctoral Poster Presentation in CS*
- [O2] Belov, A., Belyaev, P., Viksnin, I., Kim, I., Radabolsky, V., Turushev, T., & **Chuprov, S.** (2023). “Software and Hardware Bundle for Controlling a Group of Unmanned Vehicles Based on Robust Computer Vision Algorithms” in *LETI Transactions on Electrical Engineering & Computer Science*. 2023, vol. 16, no. 6. pp. 52–69. – *in Russian*. [URL](#)
- [O3] Berezovskaya, O., **Chuprov, S.**, Neverov, E., & Sadreev., E. (2022). “Review and Comparison of AES Lightweight Modifications for a Low-Power Devices Network” in *Problems of Information Security. Computer Systems*. no. 2, pp. 35-50. doi: 10.48612/jisp/gp9v-96dh-32 v1. – *in Russian*. [URL](#)
- [O4] Buzina, E., **Chuprov, S.**, Tursukov, N., Belyaev, P., & Viksnin, I. (2022). “Algorithm for Uniform Coverage of the Monitoring Territory by Unmanned Aerial Vehicles” in *LETI Transactions on Electrical Engineering & Computer Science*. 2022. Vol. 15, no. 5/6. P. 41–50. doi: 10.32603/2071-8985-2022-15-5/6-41-50. – *in Russian*. [URL](#)
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- [O6] **Chuprov, S.** (2020). “Intersection Management Tasks Application in Mobile Robotic System with Autonomous Vehicle Models” in *Saint-Petersburg’s Government Grant Award Winners*, pp. 68-74. [URL](#)
- [O7] **Chuprov, S.** (2020). “Reputation, Trust, and Data Quality Concept for Security and Safety Assurance in a Group of Autonomous Vehicles” in *IX Young Scientists Congress Proceedings, Electronic Edition*. [URL](#)
- [O8] **Chuprov, S.** (2019). “Security Assurance of Mobile Robotic Systems with Decentralized Control” in *2019 ITMO University Annotated Master’s Research Thesis Proceedings*, pp. 66-72. [URL](#)

- [O9] **Chuprov, S.**, Viksnin, I., & Kim, I. (2019). “Organization of Safe Intersections Passage by Unmanned Vehicles in a Mobile Robotic System” in VIII Young Scientists Congress Proceedings, Electronic Edition. [URL](#)
- [O10] **Chuprov, S.**, Viksnin, I., Kim, I., & Usova, M. (2019). “Secure Intersection Management Passage with a Group of Autonomous Vehicles in Mobile Robotic System with Decentralized Control” in ITMO University Young Scientists Almanac, vol. 2, pp. 43-48. [URL](#)
- [O11] **Chuprov, S.**, Viksnin, I., & Usova, M. (2018). “Police Office Model with Partial Decentralized Control in Mobile Robotic System” in Regional Informatics and Information Security SPOISU Proceedings, vol. 6, pp. 245-249. [URL](#)

Presentations, posters, forums, workshops, and seminars w/o publications:

- [NP1] Fonseca, P., & **Chuprov, S.** (2025). “SPAM Email Detection App Using Machine Learning Techniques” in *STARTER-AI Summer 2025 Workshop*. Presented May 21, 2025.
- [NP2] **Chuprov, S.** (2024). “About my Research, Approach to Mentorship, and Some Advice from my Fresh Grad Student’s Perspective” at First-Year PhD Students in Computer Science seminar at UTRGV, 16 September 2024, Edinburg, TX, USA.
- [NP3] **Chuprov, S.** (2024). “Security and Robustness of Machine Learning under Vulnerable Cyberinfrastructure and Varying Data Quality” at [From Theory to Practice \(T2P\)](#) Workshop, 15-19 April 2024, Kigali, Rwanda – *Remote presentation*.
- [NP4] **Chuprov, S.**, Zatsarenko, R., & Reznik, L. (2024). “Improving Federated Learning Robustness towards Security Violations and Data Quality Degradations OR How to Learn Better via Extracting Knowledge and Accumulating History?” at [UPSTAT 2024](#), 12-13 April 2024, Rochester, NY, USA.
- [NP5] **Chuprov, S.**, & Reznik, L. (2023). “FLAME: Federated Learning against Malicious Engineering. Employing Trust and Reputation to Enhance Learning Security and Privacy” at Rochester Security Summit (RSS:2023), 25-26 October 2023, Rochester, NY, USA.
- [NP6] **Chuprov, S.**, Reznik, L., & Zatsarenko, R. (2023). “MLIN or How to Make Networks and ML Applications Work Together in Real Conditions?” at First IEEE Upstate New York Workshop on Secure and Sustainable Communications Networks (SSCN), 10 October 2023, Rochester Institute of Technology, Rochester, NY, USA.
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